

UNIT-III

Syntax II: Models for ambiguity Resolution in parsing. Multilingual issues

Semantic Parsing I: Introduction, Semantic Interpretation, System Paradigms, Word Sense

Ambiguity in Sentences

- **Ambiguity** happens when a sentence can be parsed in more than one valid way according to grammar rules.

Example sentence:

"I saw the man with the telescope."

- This sentence has **two possible meanings**:
 - 1.Instrumental meaning:** I used a telescope to see the man.
 - 2.Attachment meaning:** The man I saw had a telescope.

Models for Resolutions of ambiguity parsing

- Probabilistic Context Free Grammar (PCFG)
- Generative model for parsing
- Discriminative model for parsing

What is a PCFG?

- A **Probabilistic Context-Free Grammar (PCFG)** is a **CFG** where **each production rule is assigned a probability**.
 - Instead of just generating possible parse trees, PCFG helps us choose the **most likely parse**.
 - Useful because **natural language is ambiguous** (a sentence can have multiple parse trees).

- **Example PCFG**

Rules with probabilities:

$S \rightarrow NP, VP, [1.0]$

$NP \rightarrow Det, N [0.6]$

$NP \rightarrow Name [0.4]$

$VP \rightarrow V, NP [0.5]$

$VP \rightarrow V [0.5]$

$Det \rightarrow the [0.8]$

$Det \rightarrow a [0.2]$

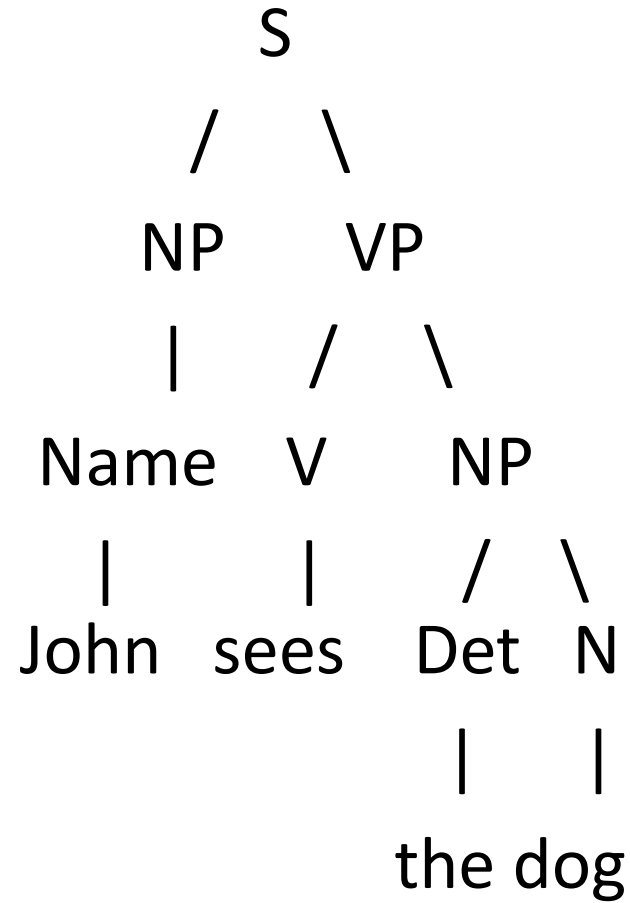
$N \rightarrow cat [0.5]$

$N \rightarrow dog [0.5]$

$Name \rightarrow John [1.0]$

$V \rightarrow sees [1.0]$

Example: John sees the dog



$$=1.0 \times 0.4 \times 1.0 \times 0.5 \times 1.0 \times 0.6 \times 0.8 \times 0.5 = 0.048$$

Rules with probabilities:

$S \rightarrow NP, VP, [1.0]$

$NP \rightarrow Det, N [0.6]$

$NP \rightarrow Name [0.4]$

$VP \rightarrow V, NP [0.5]$

$VP \rightarrow V [0.5]$

$Det \rightarrow the [0.8]$

$Det \rightarrow a [0.2]$

$N \rightarrow cat [0.5]$

$N \rightarrow dog [0.5]$

$Name \rightarrow John [1.0]$

$V \rightarrow sees [1.0]$

Production rules considered:

$S \rightarrow NP, VP = 1.0$

$NP \rightarrow Name = 0.4$

$Name \rightarrow John = 1$

$VP \rightarrow V, NP = 0.5$

$V \rightarrow sees = 1$

$NP \rightarrow Det, N = 0.6$

$Det \rightarrow the = 0.8$

$N \rightarrow dog = 0.5$

- Formal definition:

A PCFG is defined as a 5 tuple

$G=(N, \Sigma, R, S, P)$

N : Set of **non-terminals** (e.g., S, NP, VP)

Σ : Set of terminals (words, e.g., “dog”, “runs”)

$R \rightarrow$ Set of production rules (e.g., $NP \rightarrow \text{Det } N$)

$S \rightarrow$ Start symbol (e.g., Sentence: S)

$P \rightarrow$ Probability distribution over rules

Constraint:

For each non-terminal A ,

$$\sum P(A \rightarrow \beta) P(A \rightarrow \beta) = 1$$

Example:

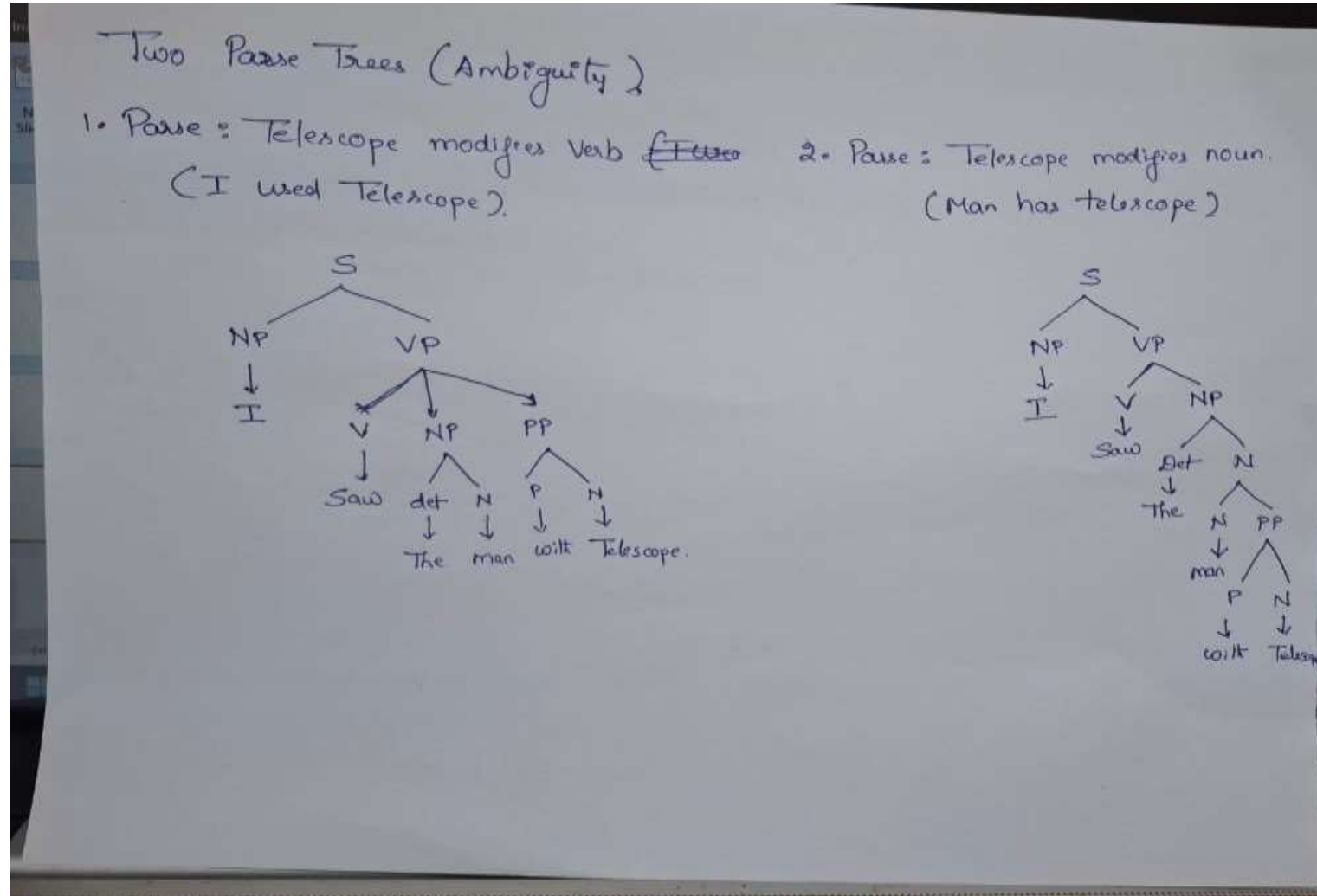
- **Example Rules (with probabilities):**

- $S \rightarrow NP, VP$ [1.0]
- $VP \rightarrow V, NP$ [0.7]
- $VP \rightarrow VP, PP$ [0.3]
- $NP \rightarrow Det, N$ [0.6]
- $NP \rightarrow N, PP$ [0.4]
- $PP \rightarrow P, N$ [1.0]
- $Det \rightarrow the$ [1.0]
- $N \rightarrow man$ [0.5]
- $N \rightarrow telescope$ [0.5]
- $NP \rightarrow I$ [1.0]
- $V \rightarrow saw$ [1.0],
- $P \rightarrow with$ [1.0]

I saw the man with the telescope

- Example sentence:

"I saw the man with the telescope."



```
import nltk
from nltk import PCFG
from nltk.parse import ViterbiParser

grammar = PCFG.fromstring("""
    S -> NP VP [1.0]
    VP -> V NP [0.5] | VP PP [0.5]
    NP -> Det N [0.3] | NP PP [0.3] | 'I' [0.2] | N [0.2]
    Det -> 'the' [1.0]
    N -> 'man' [0.5] | 'telescope' [0.5]
    V -> 'saw' [1.0]
    PP -> P NP [1.0]
    P -> 'with' [1.0]
    """)
```

```
from nltk.parse import InsideChartParser

# Sentence to parse
sentence = ['I', 'saw', 'the', 'man', 'with', 'telescope']

parser = InsideChartParser(grammar)

parses = list(parser.parse(sentence))
print("Number of parses:", len(parses))

for tree in parses:
    print(tree)
    print("Probability:", tree.prob())
    tree.pretty_print()
```

